

# Week 3

# Let's Talk About Transforms

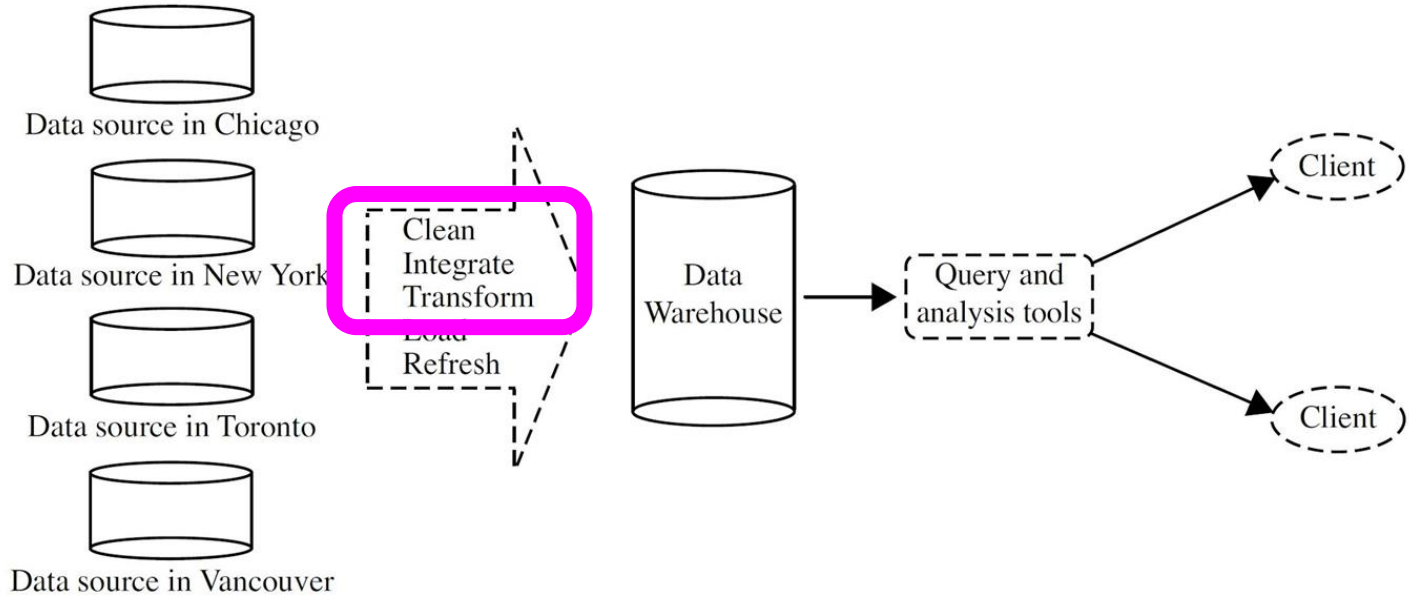
ISTA 322 - Data Engineering

# Last week - recap

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- Data types and data structures
- Python refresher
- Introduction to Google colab / Jupyter notebooks

# Middle out



**Figure 1.6** Typical framework of a data warehouse for *AllElectronics*.

# Starting with the T in ETL

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- Why?
- You all know *some* python
- Most of you have done some kind of data transforms
- It's the core of ETL
- The other ends are a bit more of making technology work for you (sorta)
- Overall, easier to build from the T in ETL

# Data transforms

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- Transforms are key to making data useful
- Two general purposes of transforms
  - Fixing errors
  - Making data useable / relevant
- We'll cover major examples within each
  - Solid foundation / framework
  - You'll need to learn more throughout your careers
  - \* Importance of “learning to learn”
- Depends a lot on existing data quality and infrastructure

# Fixing errors

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- Existing data can be filled with errors
- Lack of precision
  - Data was entered wrong
  - Missing values
  - Bad fills
- Unnecessary
  - Too many columns
  - End user doesn't actually need
  - Long mappings
- Repetitive
  - Duplication?
  - Not normalized?

# Making data useable / relevant

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- Different clients will have different needs
- Need to transform data into a format that is useful
- Business intelligence
  - Aggregate daily statistics
  - Make key metrics to display
  - Filtering
- Data science
  - Extract data from strings
  - Scale & alter units
  - Binning/grouping
  - New features
- Of course there are more, but not covering them all

# One ETL to rule them all

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- Couple notes
- Don't think you're going to set up just one ETL
- Just because you work with a DE doesn't mean you won't have to do an ETL
- ETL doesn't have to fix everything
- We're focused on a general framework here!



# Transforms - errors

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Data sources can be filled with errors

- Bad defaults might need to be fixed
  - Fill empty cells with 9999
  - Allow too wide of range on data input
- Just allowing empty cells
  - Could be fine
  - Might need to fill

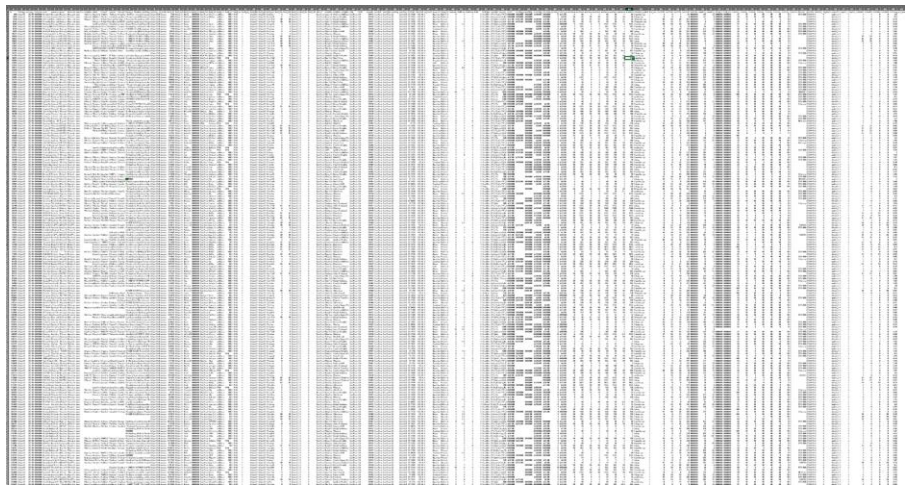
| price    | weekly_price | monthly_price | security_deposit | cleaning_fee | guests_included | extra_people | minimum_rent | maximum_rent | minimum_rent_per_week | maximum_rent_per_week | minimum_rent_per_week | maximum_rent_per_week |
|----------|--------------|---------------|------------------|--------------|-----------------|--------------|--------------|--------------|-----------------------|-----------------------|-----------------------|-----------------------|
| \$170.00 | \$1,120.00   | \$4,200.00    | \$100.00         | \$100.00     | 2               | \$25.00      | 1            | 30           | 1                     | 1                     | 30                    | 30                    |
| \$235.00 | \$1,600.00   | \$5,500.00    |                  | \$100.00     | 2               | \$0.00       | 30           | 60           | 30                    | 30                    | 60                    | 60                    |
| \$65.00  | \$485.00     | \$1,685.00    | \$200.00         | \$50.00      | 1               | \$12.00      | 32           | 60           | 32                    | 32                    | 60                    | 60                    |
| \$65.00  | \$490.00     | \$1,685.00    | \$200.00         | \$50.00      | 1               | \$12.00      | 32           | 90           | 32                    | 32                    | 90                    | 90                    |
| \$685.00 |              |               | \$0.00           | \$225.00     | 2               | \$150.00     | 4            | 1125         | 4                     | 4                     | 1125                  | 1125                  |
| \$255.00 |              |               | \$0.00           | \$125.00     | 1               | \$0.00       | 2            | 365          | 2                     | 2                     | 365                   | 365                   |
| \$139.00 |              | \$9,999.00    | \$0.00           | \$50.00      | 2               | \$60.00      | 1            | 14           | 1                     | 1                     | 14                    | 14                    |

# Transforms - errors

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Data sources can be filled with errors

- Bad defaults might need to be fixed
  - Fill empty cells with 9999
  - Allow too wide of range on data input
- Just allowing empty cells
  - Could be fine
  - Might need to fill
- Excessive data!
  - 106 columns 
- Unnecessary data
  - Remove identifying info?



| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 |

# Transforms - Making data useable / relevant

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Many ways to accomplish this

- Extract data from strings
  - Splitting columns to parse information
  - Regex fun times `'[0-9]{5}$'`
  - Data type conversions

| time                                             | userID                                | action                                          | domain             |
|--------------------------------------------------|---------------------------------------|-------------------------------------------------|--------------------|
| 2008-01-31 15:54:25<br>http://www.abc.com        | __RequestVerificationToken_Lw__=2 DB2 | ;+.ASPXAUTH=31HDWD05KU0043S                     | /product/YJ29I CVQ |
| 2005-12-02 02:36:30<br>http://www.ebay.com       | __RequestVerificationToken_Lw__=1 233 | ;+.ASPXAUTH=7HTS9Q9CC8ZX ERD/product/MVI9H HP8A |                    |
| 2015-06-02 23:27:58<br>http://www.abc.com        | __RequestVerificationToken_Lw__=B 22B | ;+.ASPXAUTH=8SZL3FPGFUS8 NA/search/P5XKO AC9    |                    |
| 2009-03-10 03:16:27<br>http://www.shophealthy.co | __RequestVerificationToken_Lw__=1 1C2 | ;+.ASPXAUTH= BWZJJR6CG85Y OM3/product/A1302 WBT |                    |

# Transforms - Making data useable / relevant

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Working with event level data

- Binning
  - 'short', 'medium', 'long' trips from trip distance
- Aggregating
  - Number of trips per time, popular locations

| VendorID | tpep_pickup_datetime | tpep_dropoff_datetime | passenger | trip_distance | Ratecode[store_and | PULocationID | DOLocationID |
|----------|----------------------|-----------------------|-----------|---------------|--------------------|--------------|--------------|
| 1        | 1/1/2018 0:21        | 1/1/2018 0:24         | 1         | 0.5           | 1 N                | 41           | 24           |
| 1        | 1/1/2018 0:44        | 1/1/2018 1:03         | 1         | 2.7           | 1 N                | 239          | 140          |
| 1        | 1/1/2018 0:08        | 1/1/2018 0:14         | 2         | 0.8           | 1 N                | 262          | 141          |
| 1        | 1/1/2018 0:20        | 1/1/2018 0:52         | 1         | 10.2          | 1 N                | 140          | 257          |
| 1        | 1/1/2018 0:09        | 1/1/2018 0:27         | 2         | 2.5           | 1 N                | 246          | 239          |
| 1        | 1/1/2018 0:29        | 1/1/2018 0:32         | 3         | 0.5           | 1 N                | 143          | 143          |

# Transforms - Making data useable / relevant

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Working with event level data

- Altering units
- Calculating useful metrics
  - Trip time
  - Speed of trip
  - Speed at different times

| VendorID | tpep_pickup_datetime | tpep_dropoff_datetime | passenger | trip_distance | Ratecode[store_and | PULocationID | DOLocationID |
|----------|----------------------|-----------------------|-----------|---------------|--------------------|--------------|--------------|
| 1        | 1/1/2018 0:21        | 1/1/2018 0:24         | 1         | 0.5           | 1 N                | 41           | 24           |
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| 1        | 1/1/2018 0:09        | 1/1/2018 0:27         | 2         | 2.5           | 1 N                | 246          | 239          |
| 1        | 1/1/2018 0:29        | 1/1/2018 0:32         | 3         | 0.5           | 1 N                | 143          | 143          |

# Checking your data

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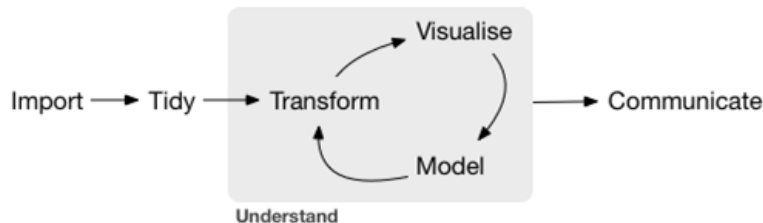
You should constantly test the ‘integrity’ of your data by asking questions and checking expectations

- General external formats
  - Right number of digits in zip/phone/social/ID/etc
- Bounded in reality
  - Age in 0-115
  - Webpage had more views than clicks
  - Only positive for some measures
- Appropriate data structure
  - Right number of rows and columns
  - Correct data types

# Wrapping up

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- The actual actions depend on the data **and** what you're going to use it for
- You'll still need to do a lot of this even if you're not a DE
  - Lots of my other DS classes have data cleaning lessons!



- Don't be afraid to use a subset to work on
- Let's go actually do this

# Tasks for this week

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- Jupyter notebook
  - Data Transforms